Medical image analysis: localising implanted electrodes in post-operative radiographs

Main PhD project • SOUND Lab, University of Cambridge • 2022–2026 • MATLAB

Linked: PhD thesis, Chapter 3

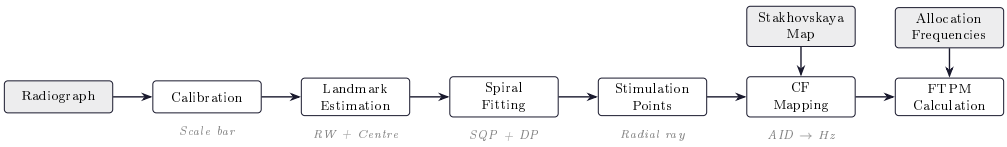
Where inside the cochlea does each implanted electrode actually sit, and how far is that from the frequency the processor sends to it?

The problem

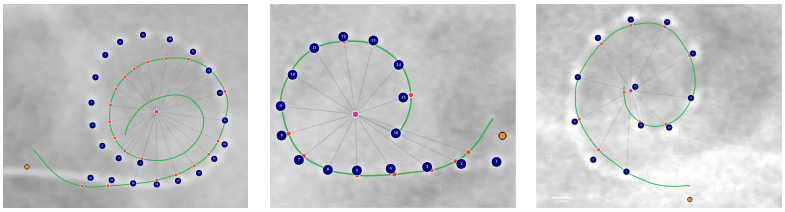
A cochlear implant assigns a frequency band to each electrode, but the place that electrode actually stimulates depends on insertion depth, which varies between patients and cannot be read off the device settings - it has to be recovered from the patient's own post-operative imaging. Clinical radiographs are 2D projections of a 3D spiral, acquired at variable magnification and orientation, so the geometry must be reconstructed rather than measured.

The pipeline

Starting from an existing semi-automatic tool (Roberts *et al.*), I refined and automated it into a pipeline that recovers per-electrode anatomical position from a single post-operative radiograph and runs reproducibly across a cohort, and I used it as the imaging basis for the thesis. Each image is calibrated from its scale bar to correct for variable magnification; electrode contacts are identified and the modiolar axis estimated by Kåsa least-squares circle fitting. A mean cochlear spiral template derived from micro-CT is registered to the electrode positions by optimising three rotation parameters under sequential quadratic programming, with electrode-to-spiral assignment solved by dynamic programming to enforce monotonic ordering and prevent turn-jumping. The fit is repeated over candidate round-window positions and scales, keeping the minimum-cost configuration. Stimulation sites follow by radial projection, are converted to angular insertion depth relative to the round window, and mapped to characteristic frequency through the Stakhovskaya spiral-ganglion function; the difference from the processor's allocated centre frequency gives a per-electrode mismatch in octaves.



From radiograph to per-electrode frequency mismatch. RW = round window, SQP = sequential quadratic programming, DP = dynamic programming, AID = angular insertion depth, CF = characteristic frequency.



Three fitted cases. Electrode contacts (blue) are matched to stimulation sites (red) on the registered cochlear spiral (green) by radial projection from the estimated modiolar axis (magenta); the round window (orange) defines the 0° reference. Insertion depth, array design and image quality differ across the three.